



Topics for Today

0. Gradients and Sharpening: Recap

I. Linear Shift Invariant Systems

II. Non-Linear Filters

III. Filtering under noise

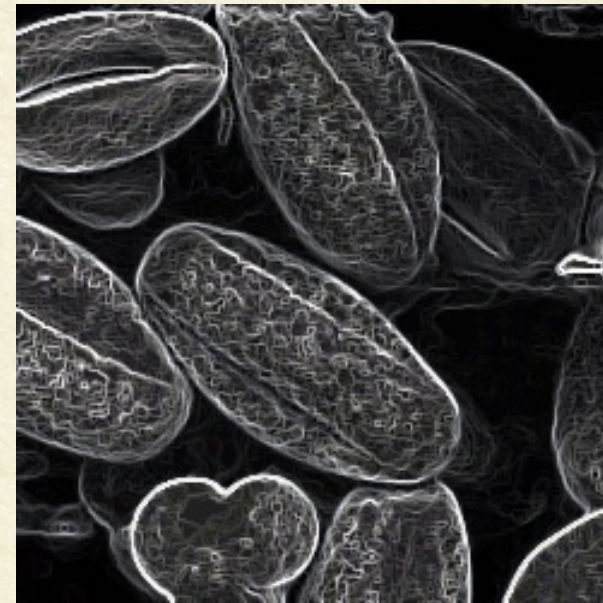
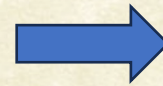
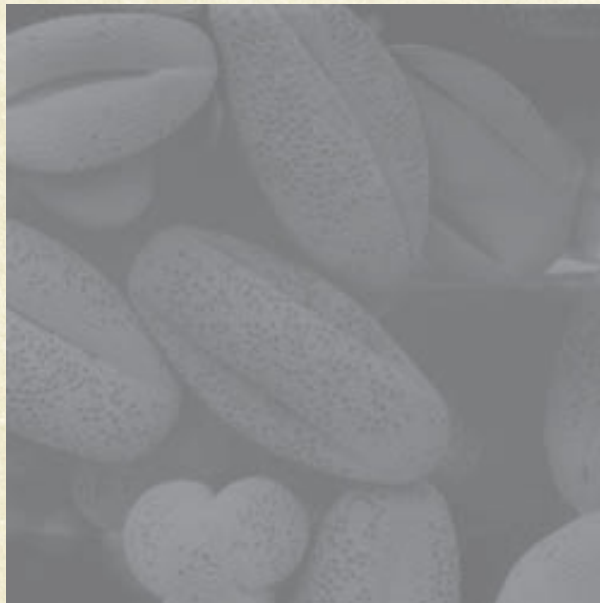
a) Bilateral Filtering

b) LoG and DoG Filters



CS7.404: Digital Image Processing

Monsoon 2025: **0**: Gradients, Sharpening: Recap



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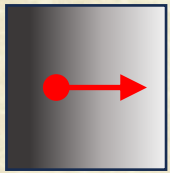


Image Gradient

The gradient of an image:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

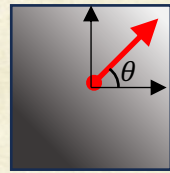
The gradient points in the direction of most rapid change (increase) in intensity.



$$\nabla f = \left[\frac{\partial f}{\partial x}, 0 \right]$$

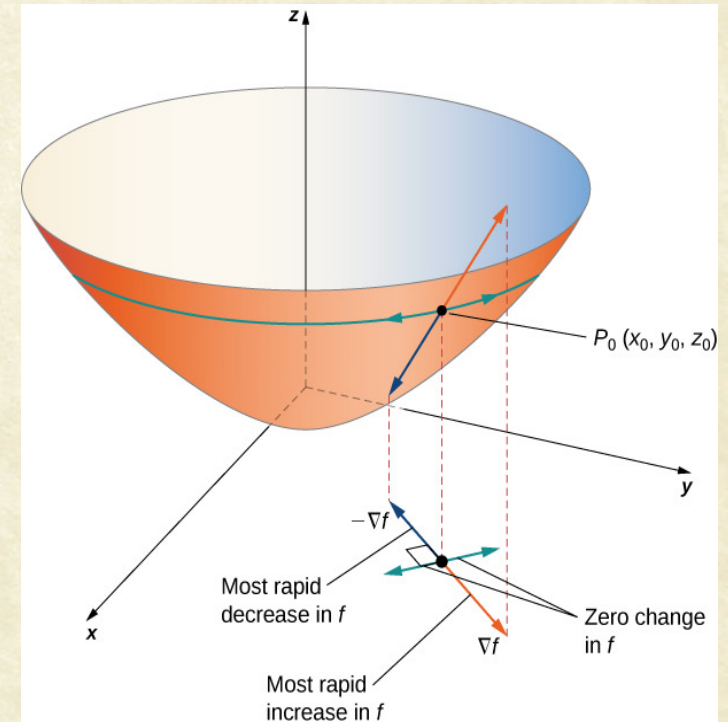
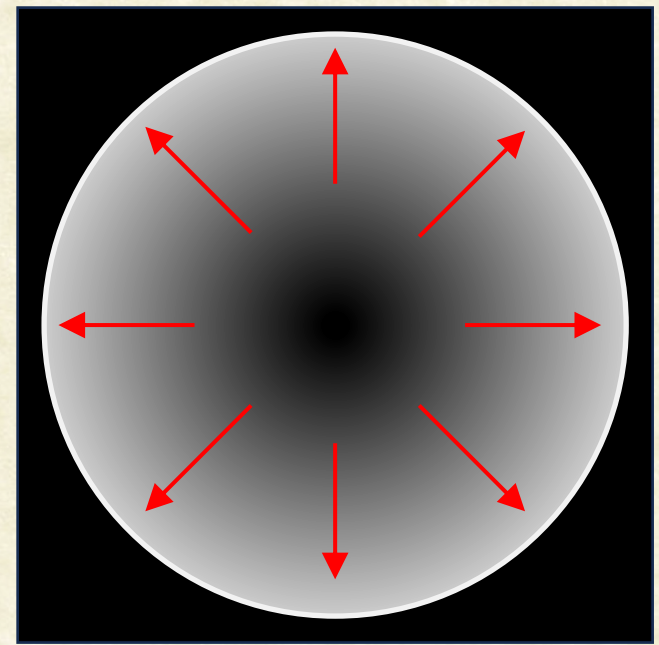


$$\nabla f = \left[0, \frac{\partial f}{\partial y} \right]$$



$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

$$\theta = \tan^{-1} \left(\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial x}} \right)$$

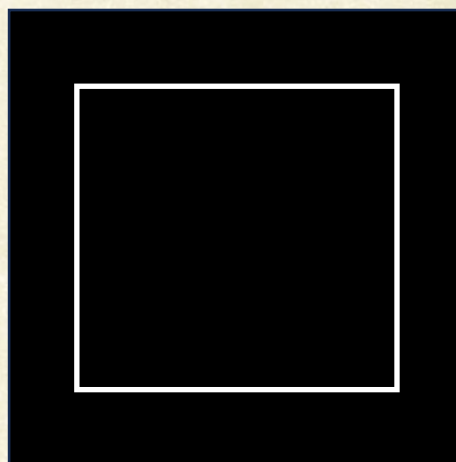




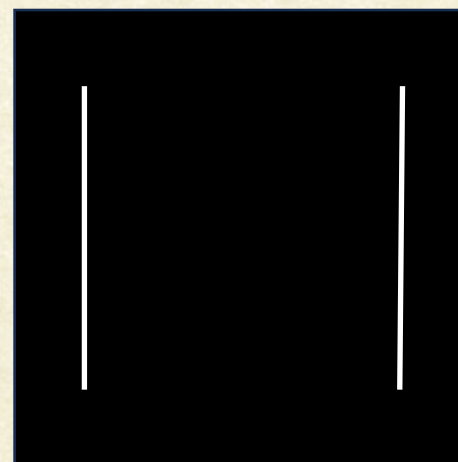
Edge is Perpendicular to Gradient



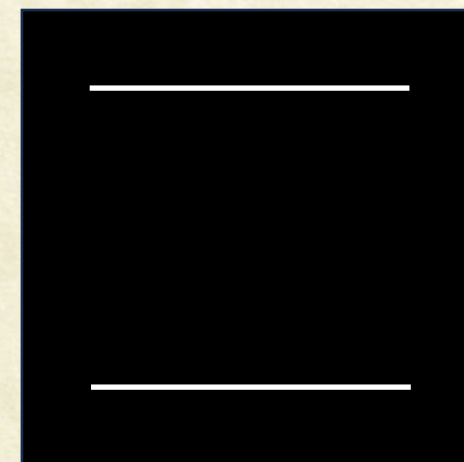
Dr. Judith Prewitt



Image



Gradient in x



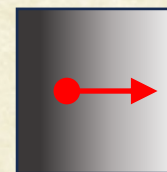
Gradient in y

-1	0	+1
-1	0	+1
-1	0	+1

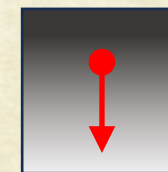
G_x

+1	+1	+1
0	0	0
-1	-1	-1

G_y



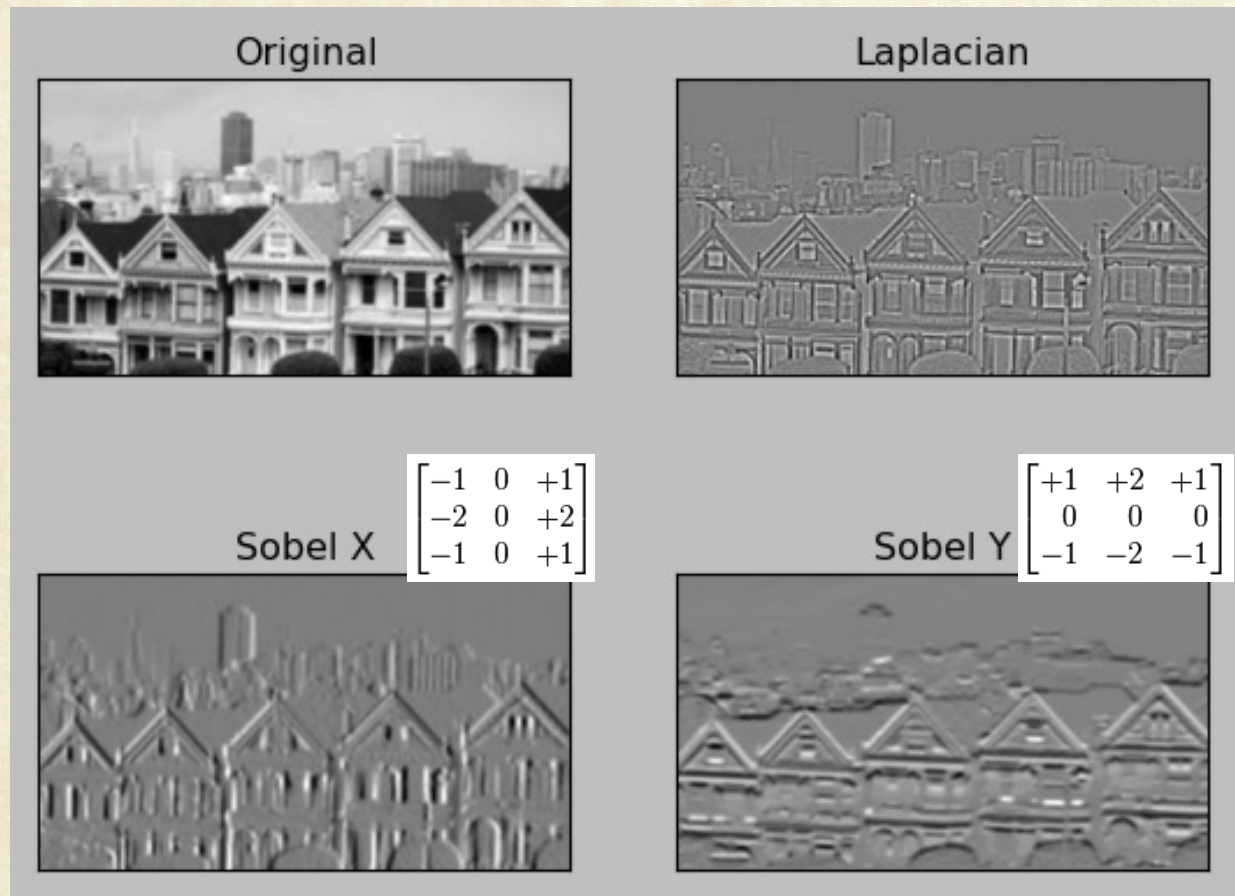
$$\nabla f = \left[\frac{\partial f}{\partial x}, 0 \right]$$



$$\nabla f = \left[0, \frac{\partial f}{\partial y} \right]$$

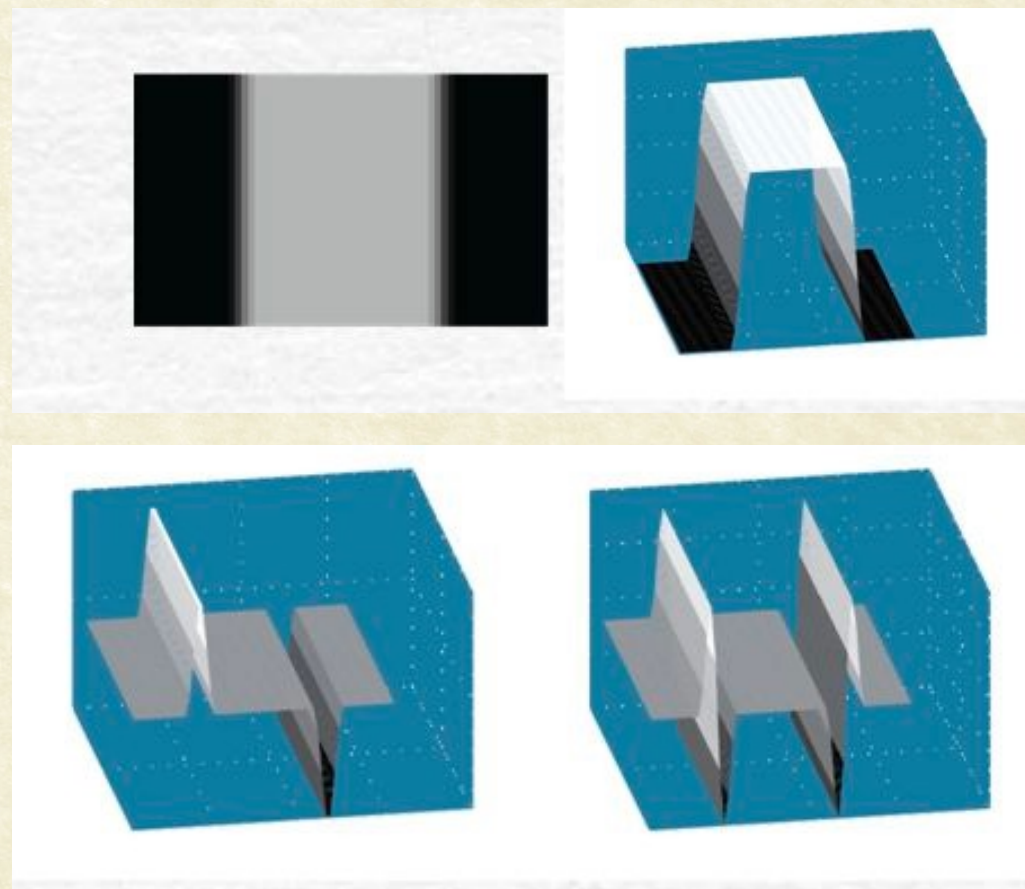


Edge Masks: Sobel, Laplacian



Note:
Coefficients
sum to 0

0	1	0
1	-4	1
0	1	0





Sharpening (Unsharp Masking)

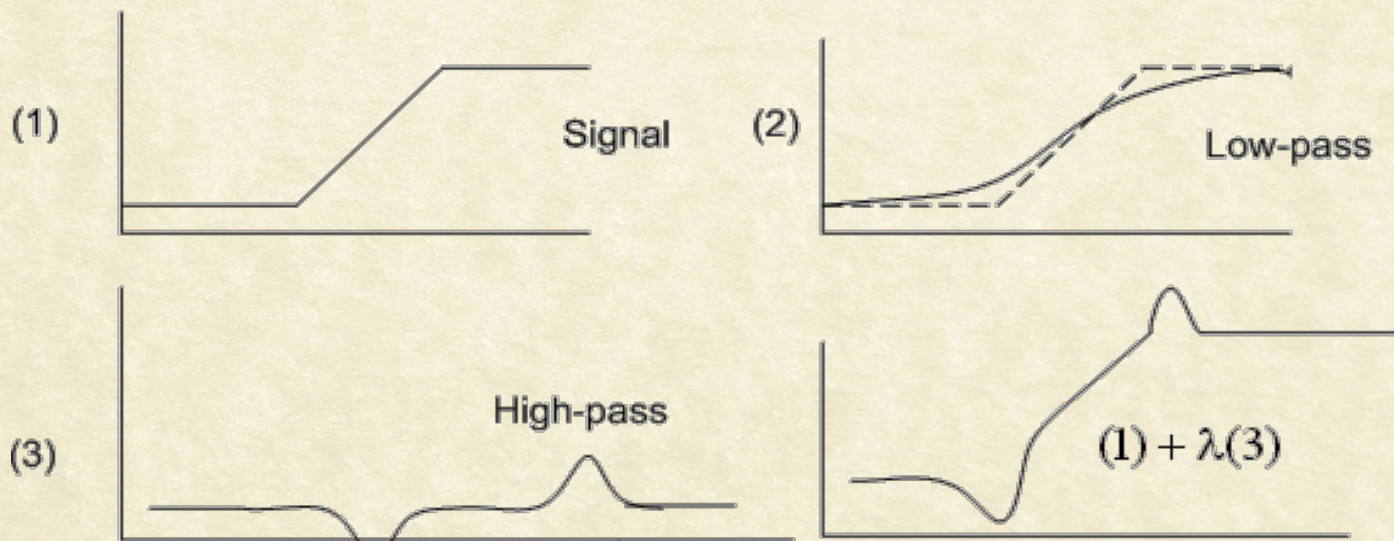
$$I(u, v)$$



$$\nabla I(u, v)$$



$$I'(u, v) = I(u, v) + \nabla^2 I(u, v)$$



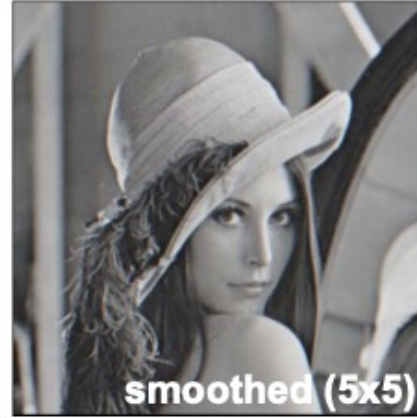


High-boost Filtering

- What does blurring take away?



—



=



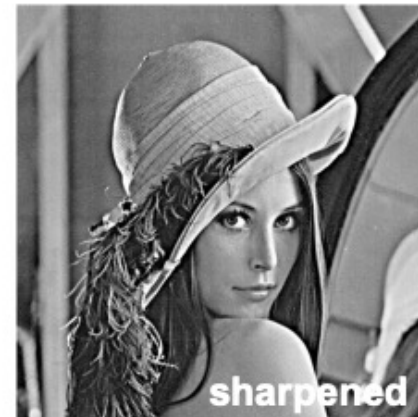
- Let's add it back:



+ a



=



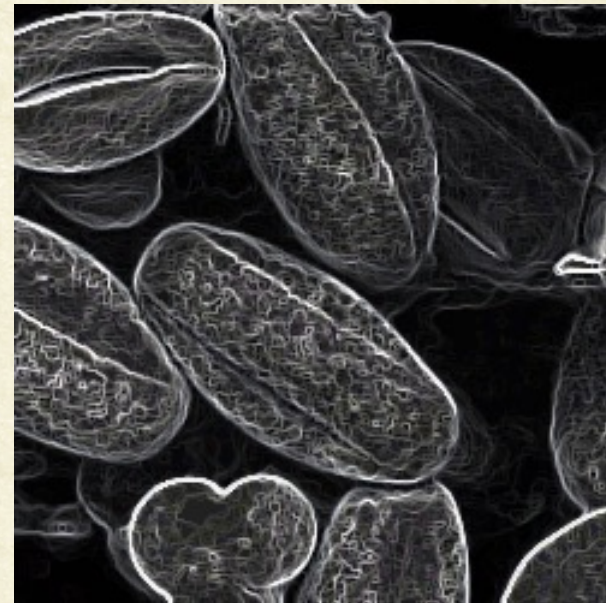
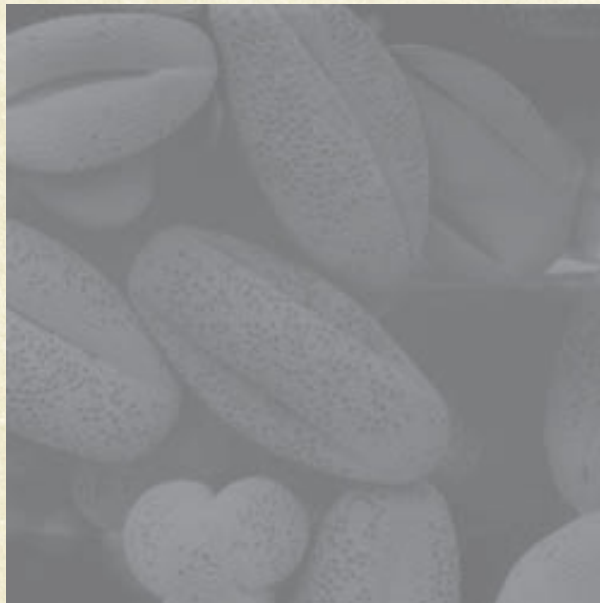


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: **I**: Linear Shift Invariant Systems

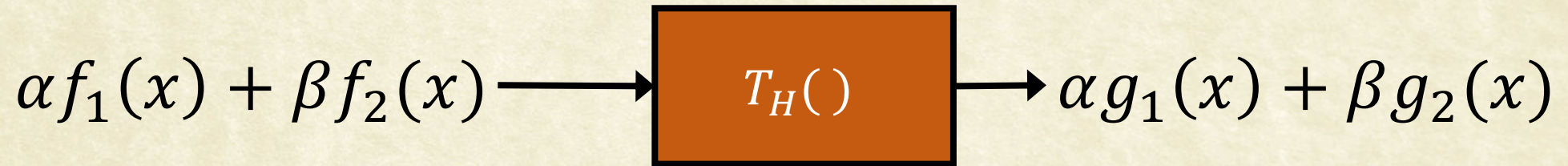
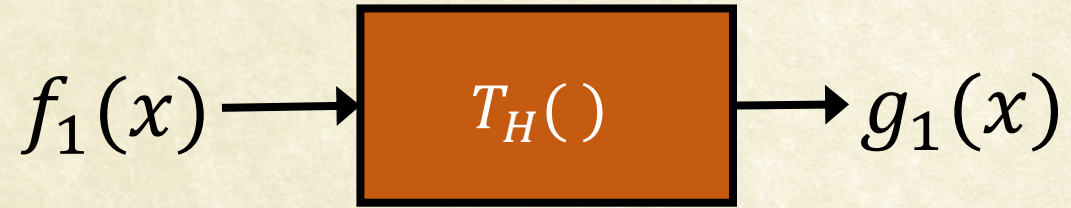


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Linear Systems

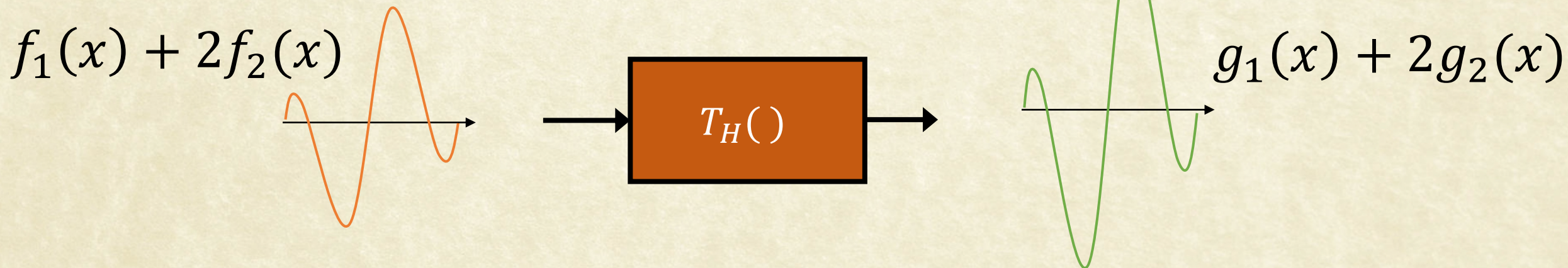
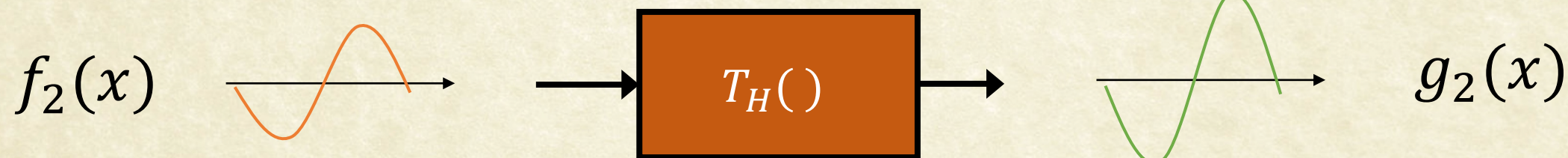
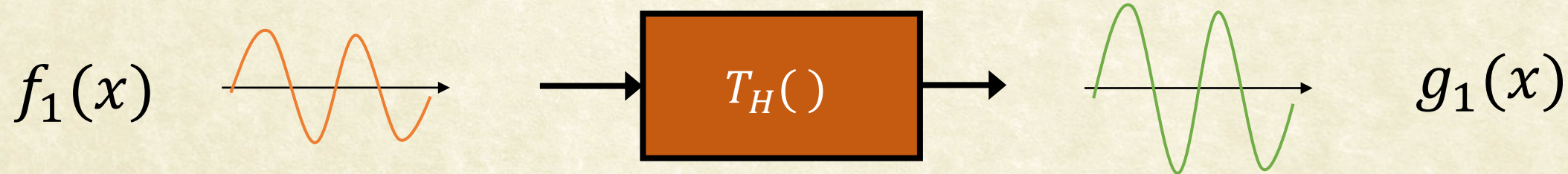
$$g(x) = T_H(f(x))$$





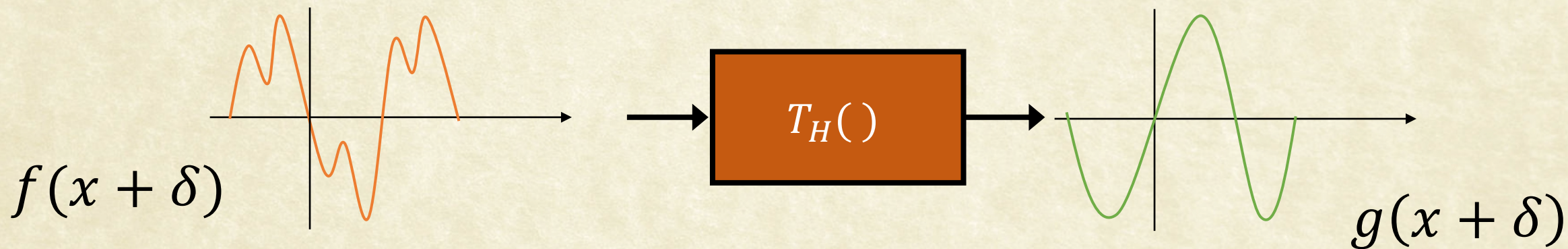
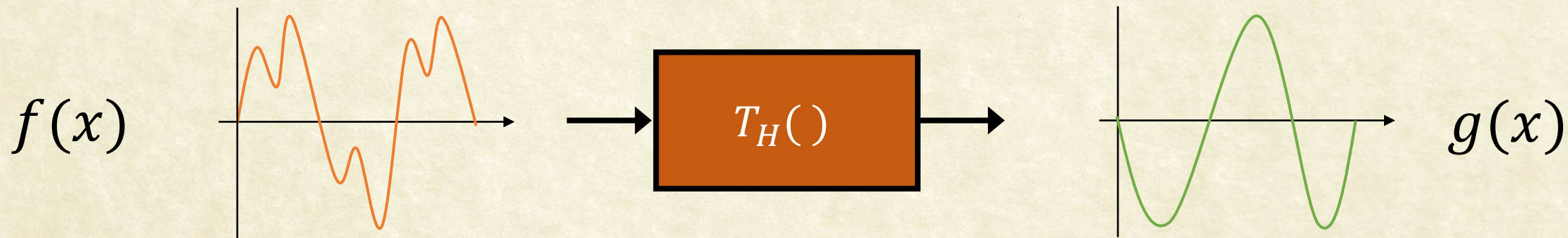
Linear Systems

$$g(x) = T_H(f(x))$$





Shift Invariant Systems





Linear Shift Invariant Systems

- Systems that are both Linear and Shift Invariant
- All convolution operations are Linear Shift Invariant
- All Linear Shift Invariant Systems can be expressed as Convolutions
- Characterized by Impulse Response or PSF
- Additional Properties
 - Separability

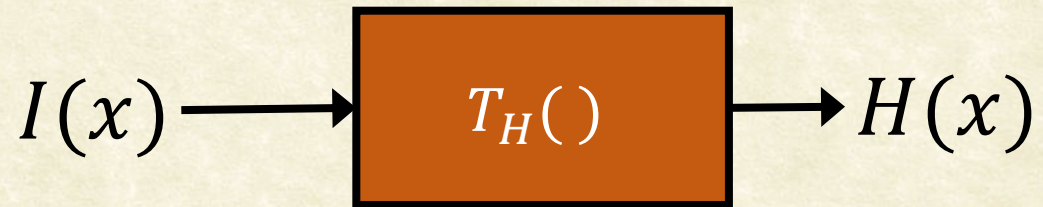
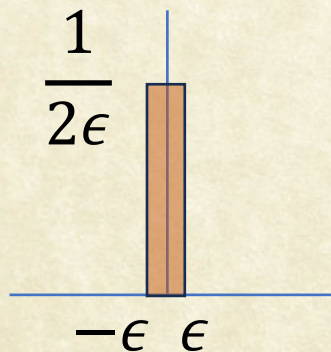


Impulse Response in Practice





Impulse Response in Practice



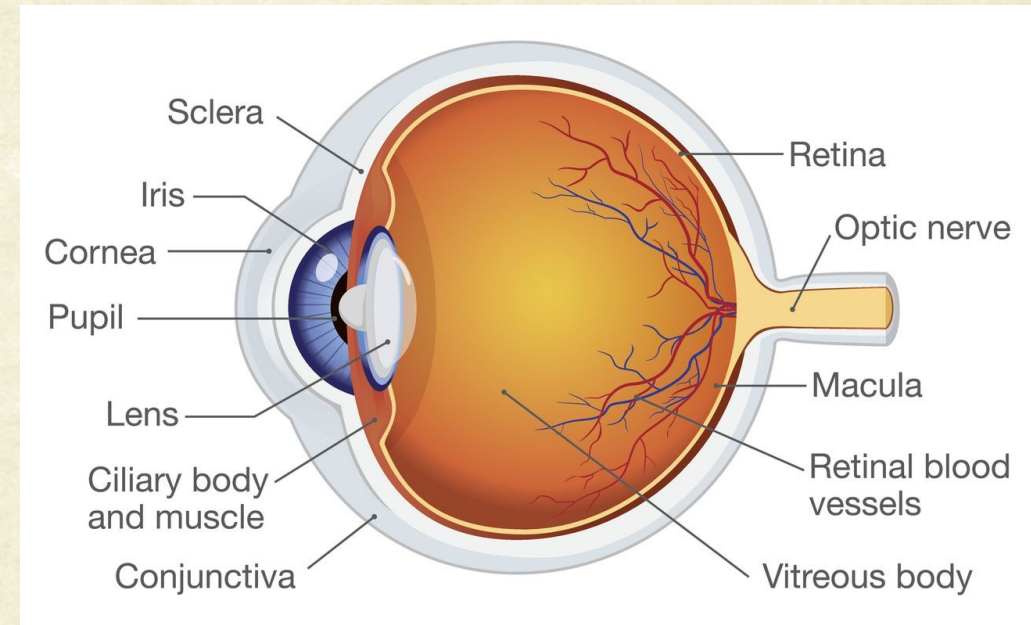
: IR or PSF





Impulse Response of Human Eye

- Input: Point Light Source ?
 - A bright (distant) star
- System: Cornea, Eye Lens, Vitreous, Retina
- Output: PSF





Homework

1. Show that Gaussian Convolutions are both LSI and Separable
2. Show that Impulse Response can fully characterize a LSIS

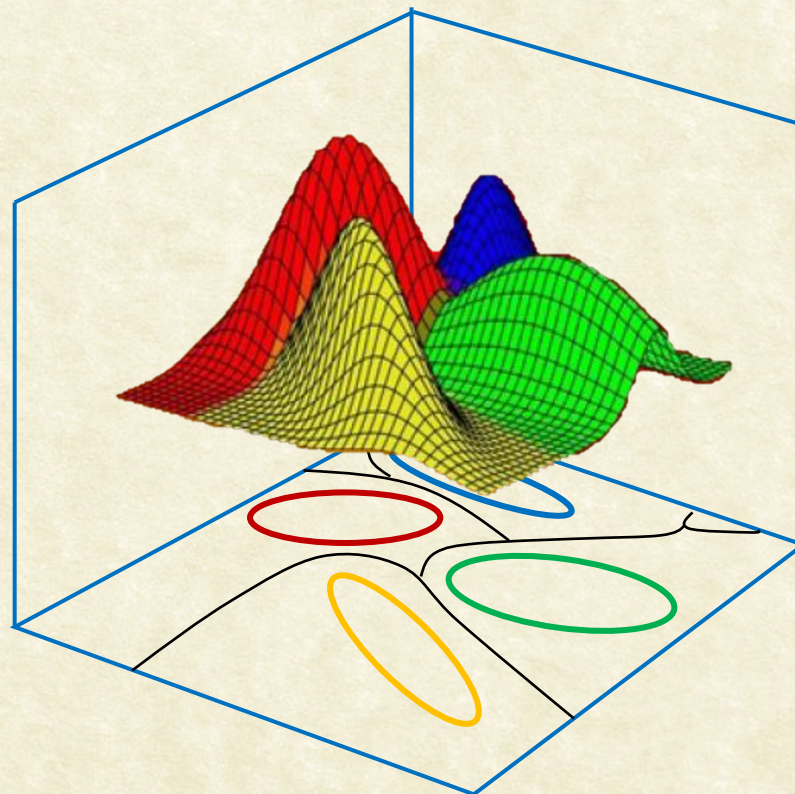


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: II: **Non-Linear Filters**



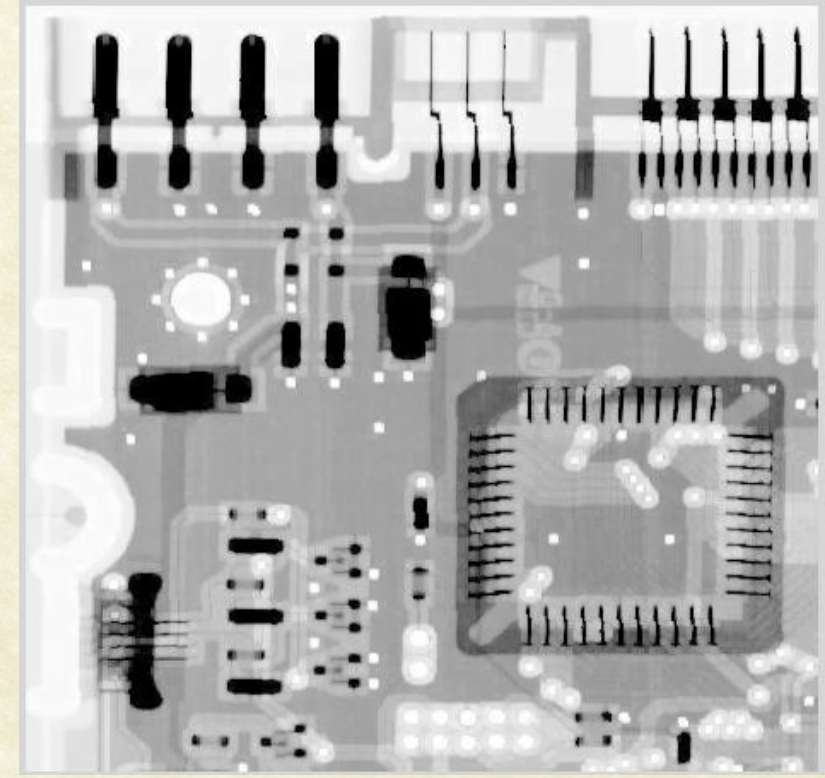
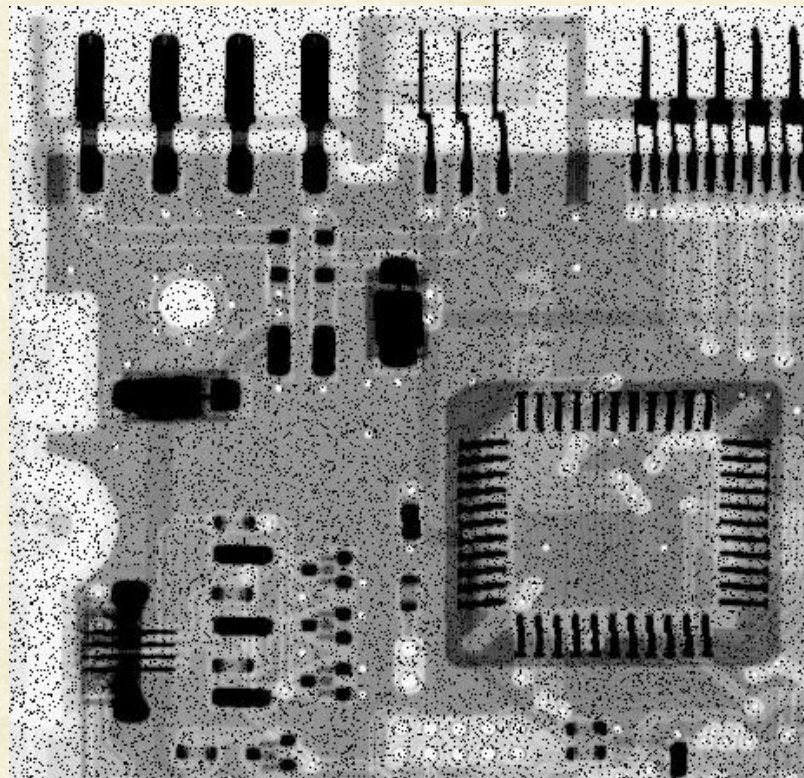
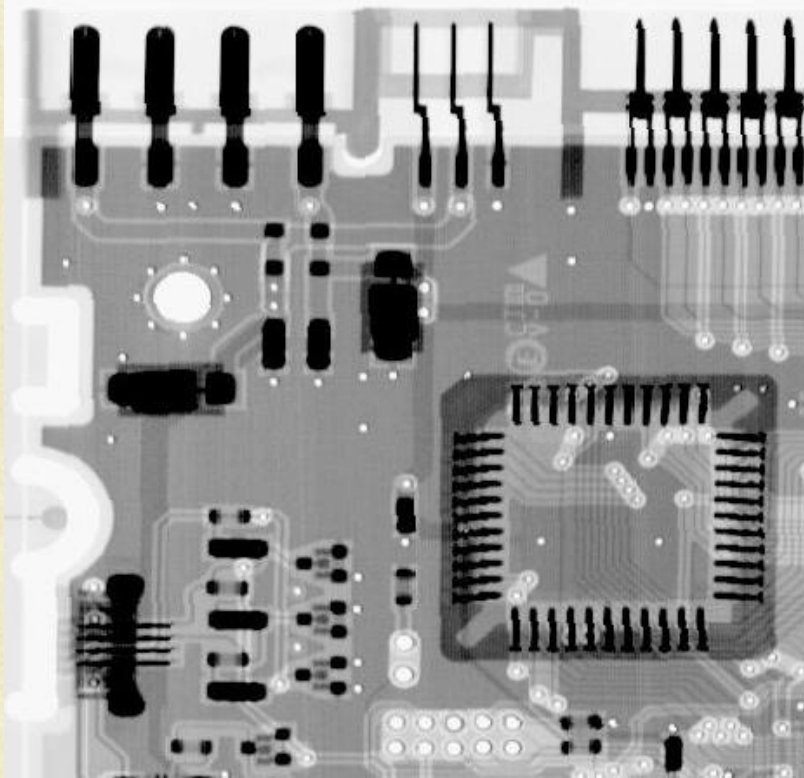
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Non-linear Spatial Filters (max)

Pepper noise

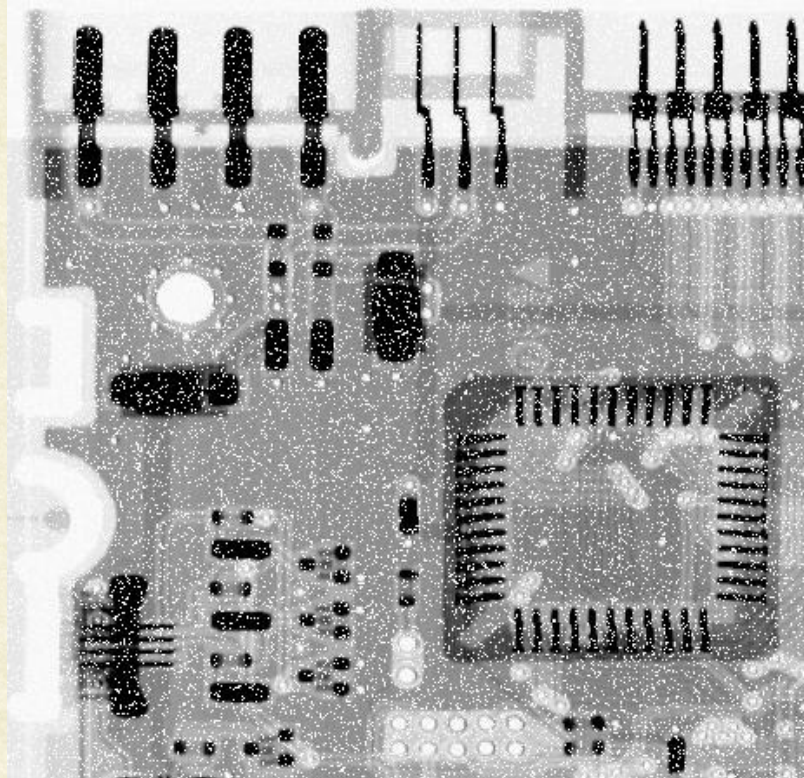
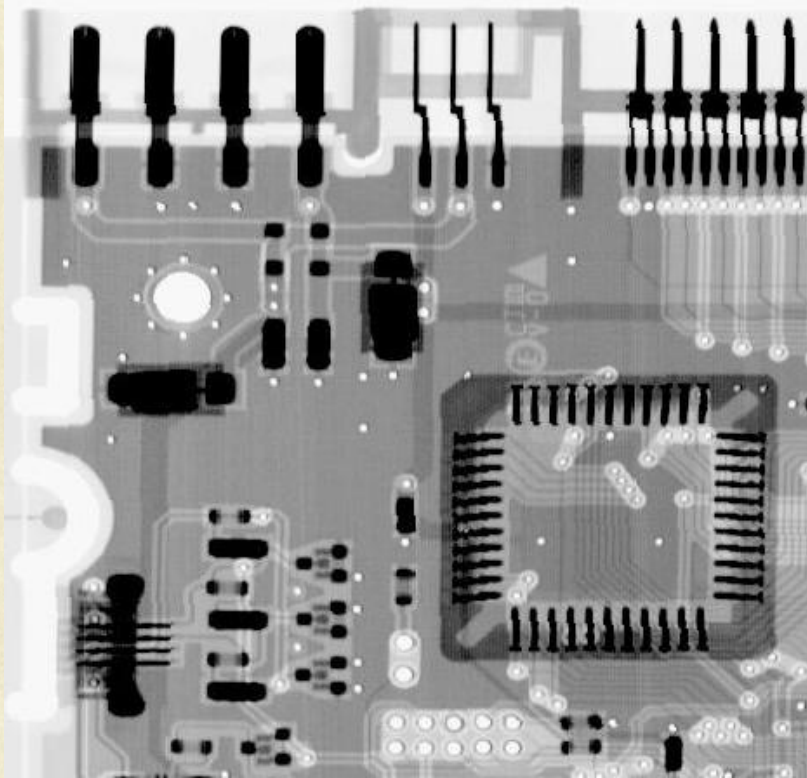
After applying max filter



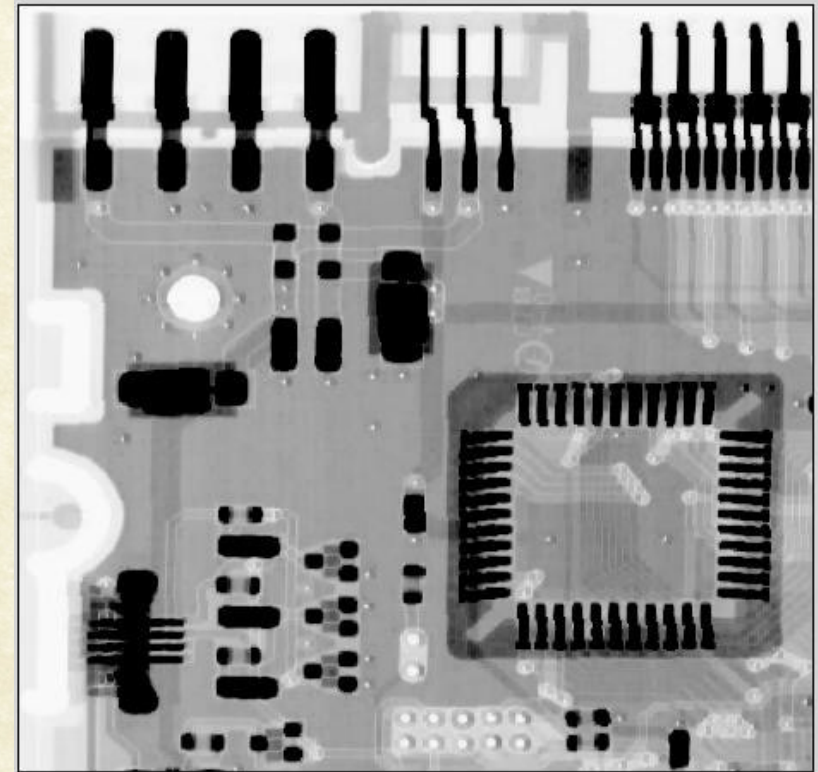


Non-linear Spatial Filters (min)

Salt noise



After applying min filter

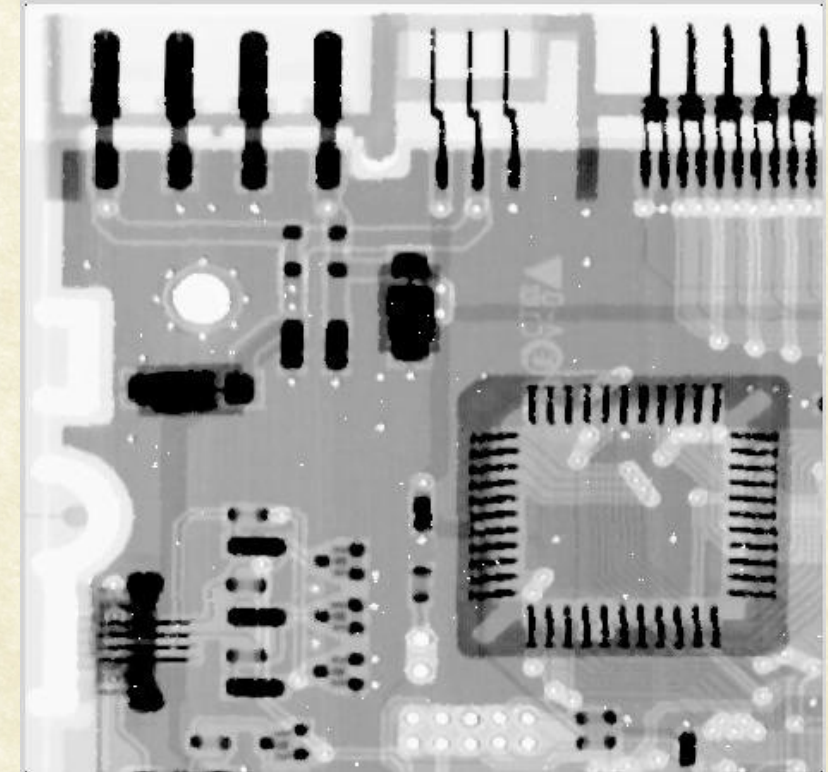
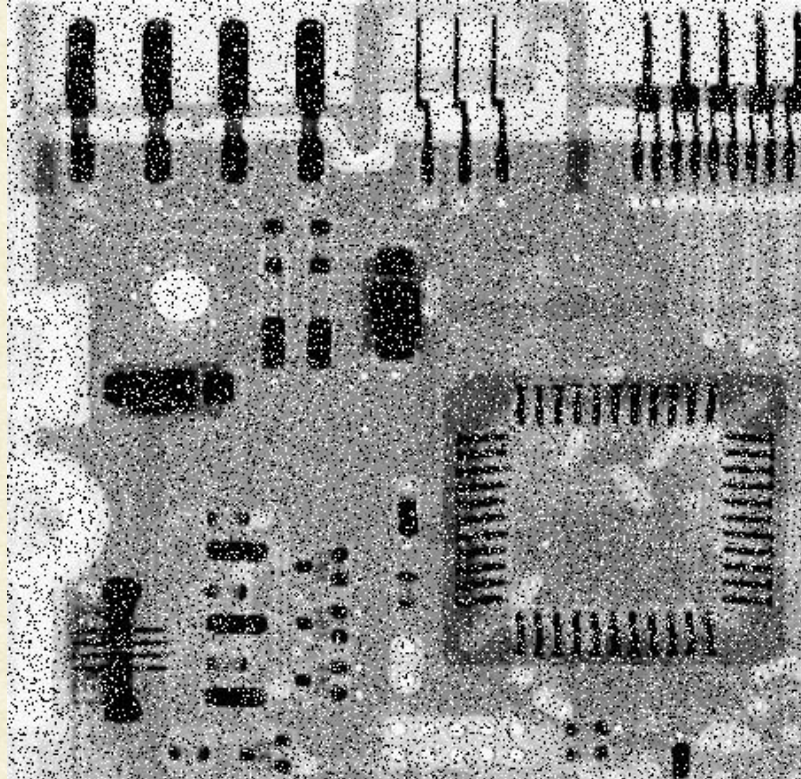
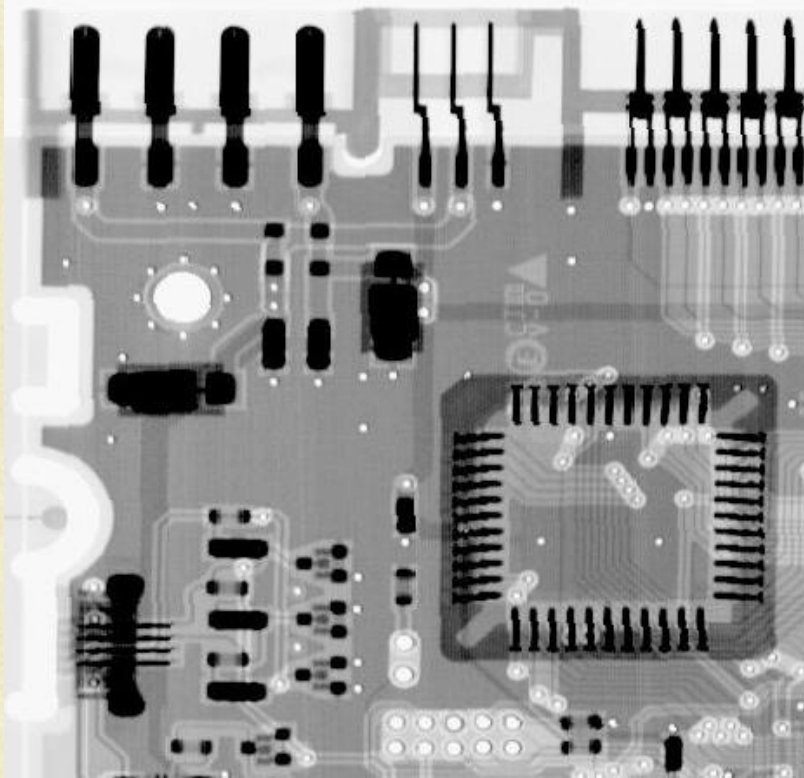




Non-linear Spatial Filters (median)

Salt & Pepper noise

After applying median filter



max, min, median → also known as rank / order statistic filters



Other Spatial Filters

- ▶ Geometric mean
- ▶ Harmonic mean
- ▶ Contra harmonic mean
- ▶ Mid Point filter
- ▶ Alpha trimmed mean filter
- ▶

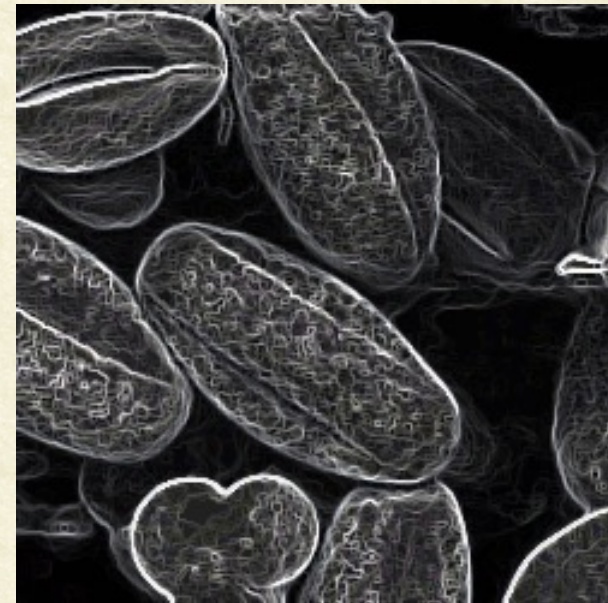
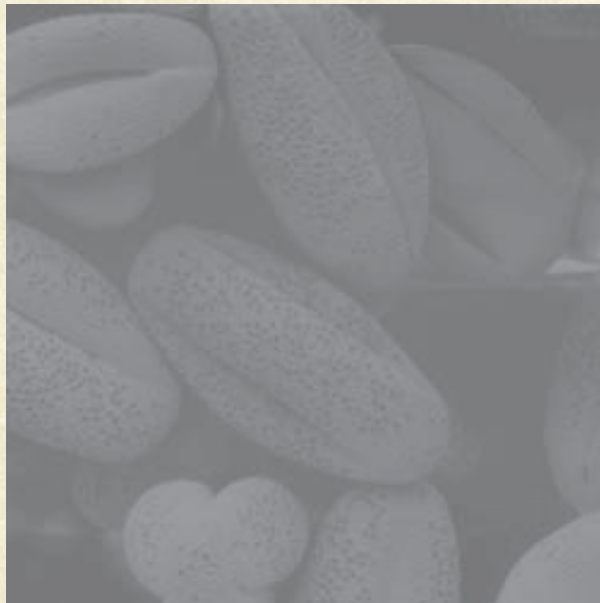


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: **III**: Filtering under Noise



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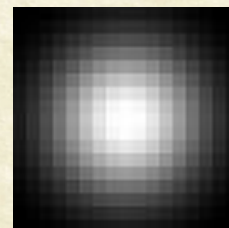
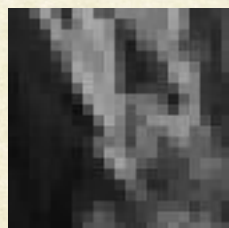
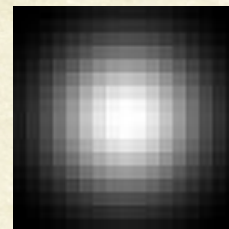
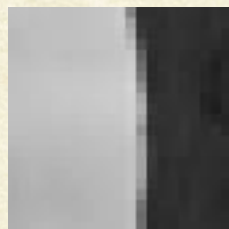
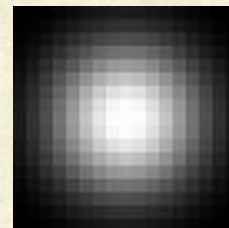
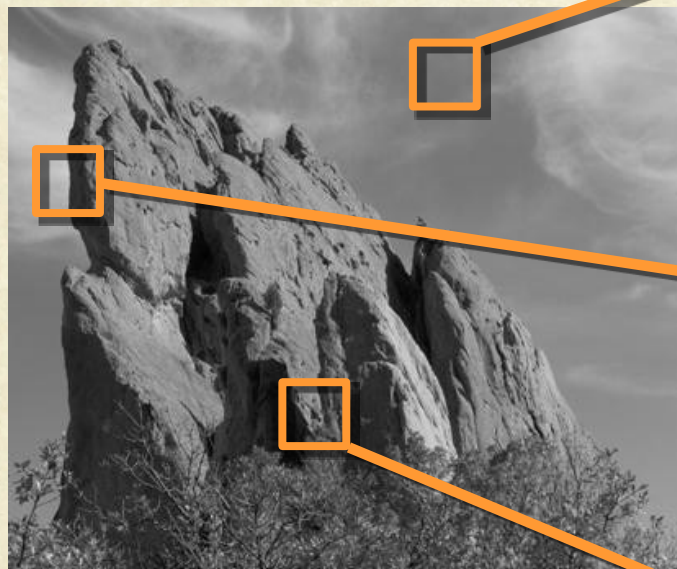
Bilateral Filtering (Edge preserving smoothing)





Gaussian Filtering

input



output

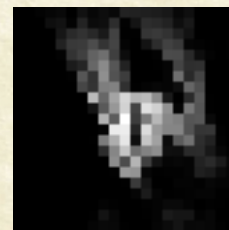
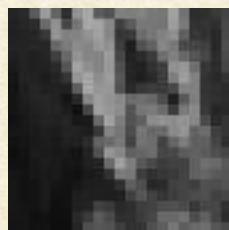
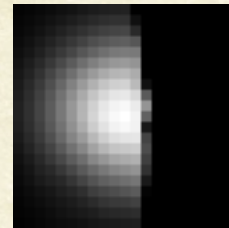
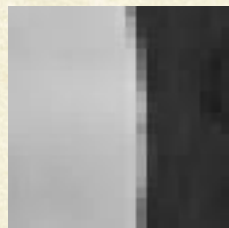
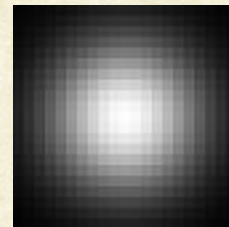
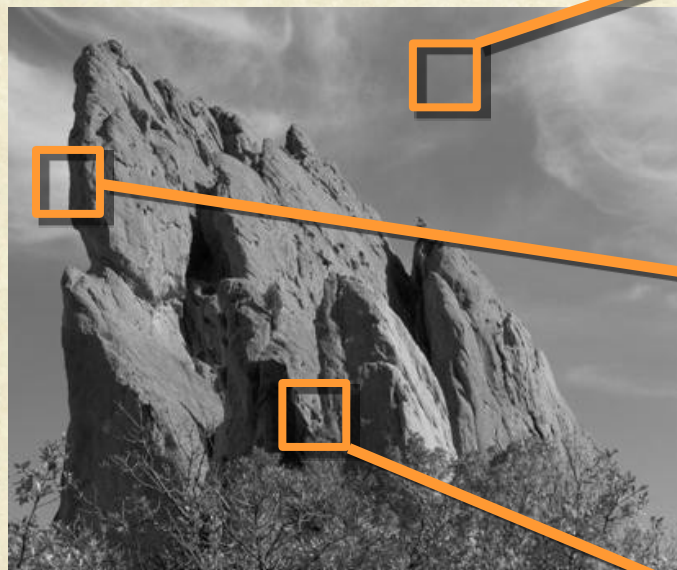


Same Gaussian kernel everywhere.

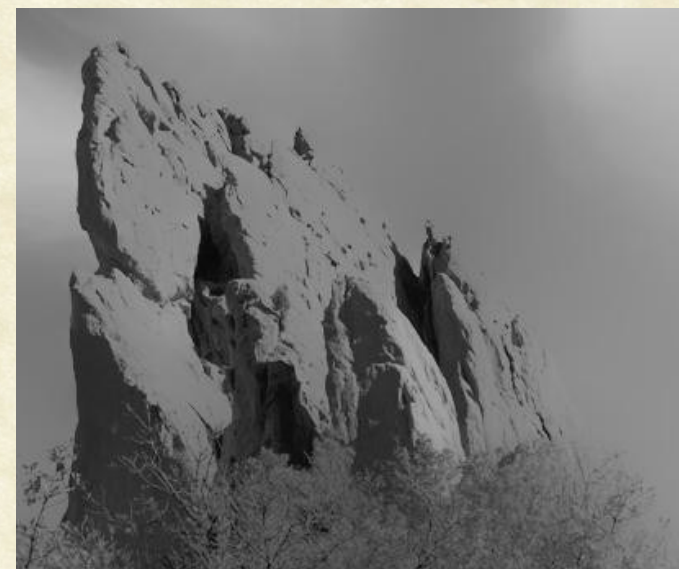


Bilateral Filtering

input



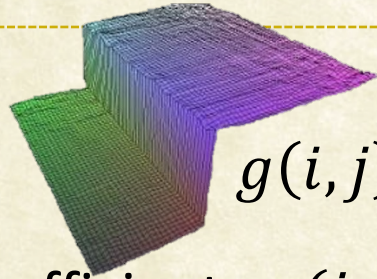
output



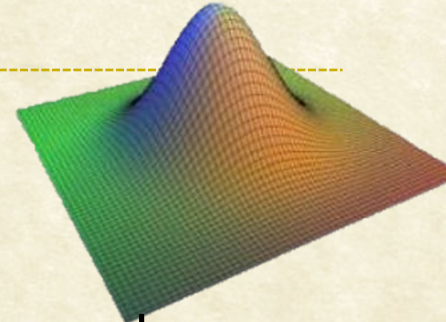
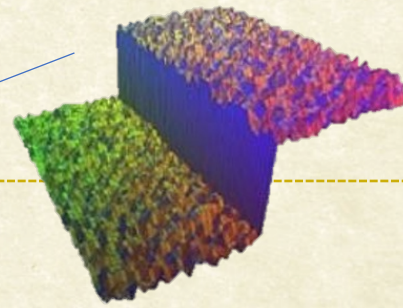
The kernel shape depends on the image content.



Bilateral Filter



$$g(i, j) = \frac{\sum_{k,l} f(k, l) \cdot w(i, j, k, l)}{\sum_{k,l} w(i, j, k, l)}$$

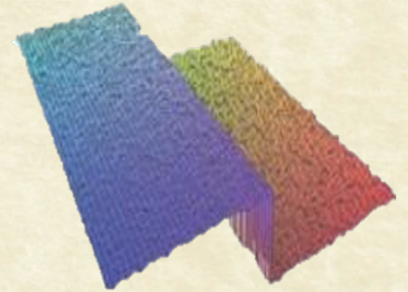


The weighting coefficient, $w(i, j, k, l)$ depends on the product of a domain kernel

$$d(i, j, k, l) = \exp\left(-\frac{(i - k)^2 + (j - l)^2}{2\sigma_d^2}\right),$$

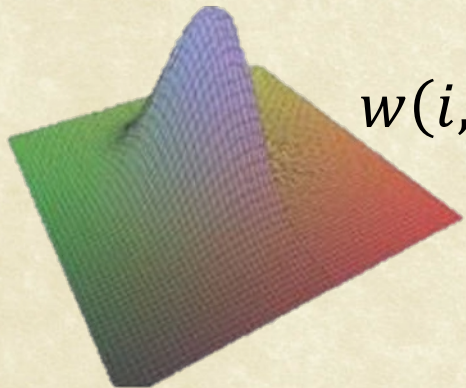
and a data-dependent range kernel

$$r(i, j, k, l) = \exp\left(-\frac{\|f(i, j) - f(k, l)\|^2}{2\sigma_r^2}\right),$$



When multiplied together, these yield the data-dependent bilateral weight function

$$w(i, j, k, l) = \exp\left(-\frac{(i - k)^2 + (j - l)^2}{2\sigma_d^2} - \frac{\|f(i, j) - f(k, l)\|^2}{2\sigma_r^2}\right)$$





$$w(i, j, k, l) = \exp \left(-\frac{(i - k)^2 + (j - l)^2}{2\sigma_d^2} - \frac{\|f(i, j) - f(k, l)\|^2}{2\sigma_r^2} \right)$$

 $\sigma_d \setminus \sigma_r$

0.05

0.2

0.8

4



8



16



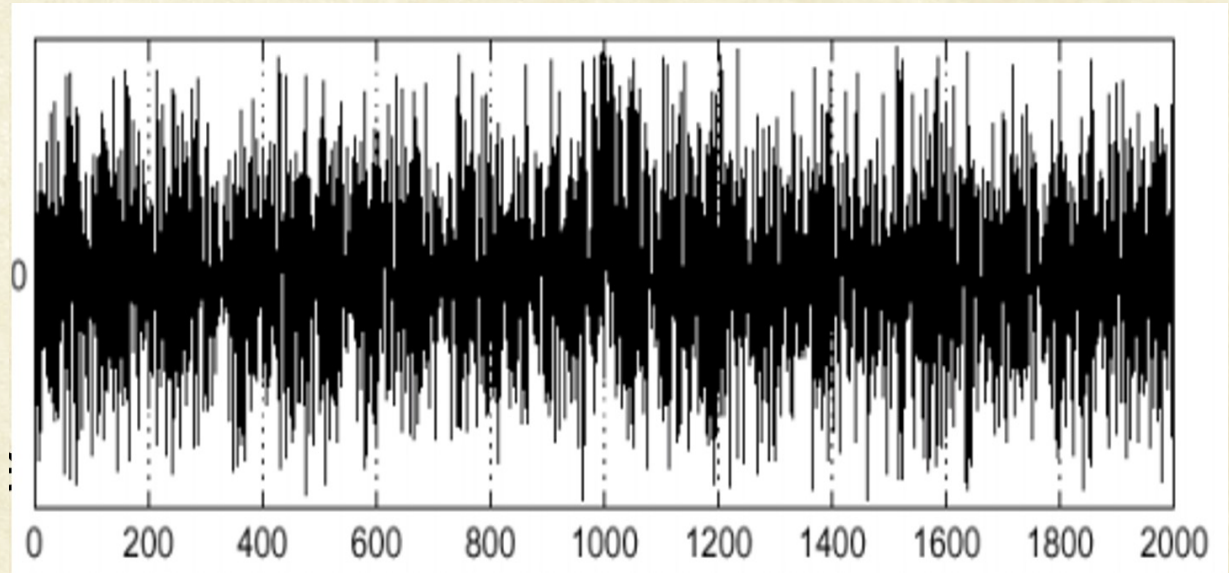
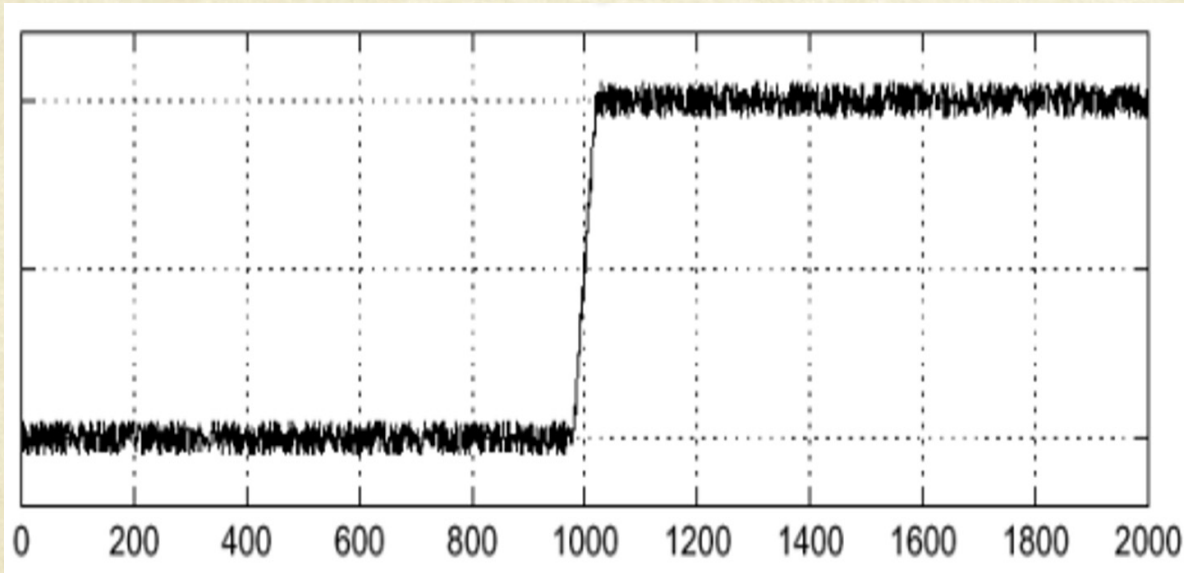


Edge Detection under Noise

- Consider a single row or column of the image
 - Plotting intensity as a function of position gives the following signal

$$f(x)$$

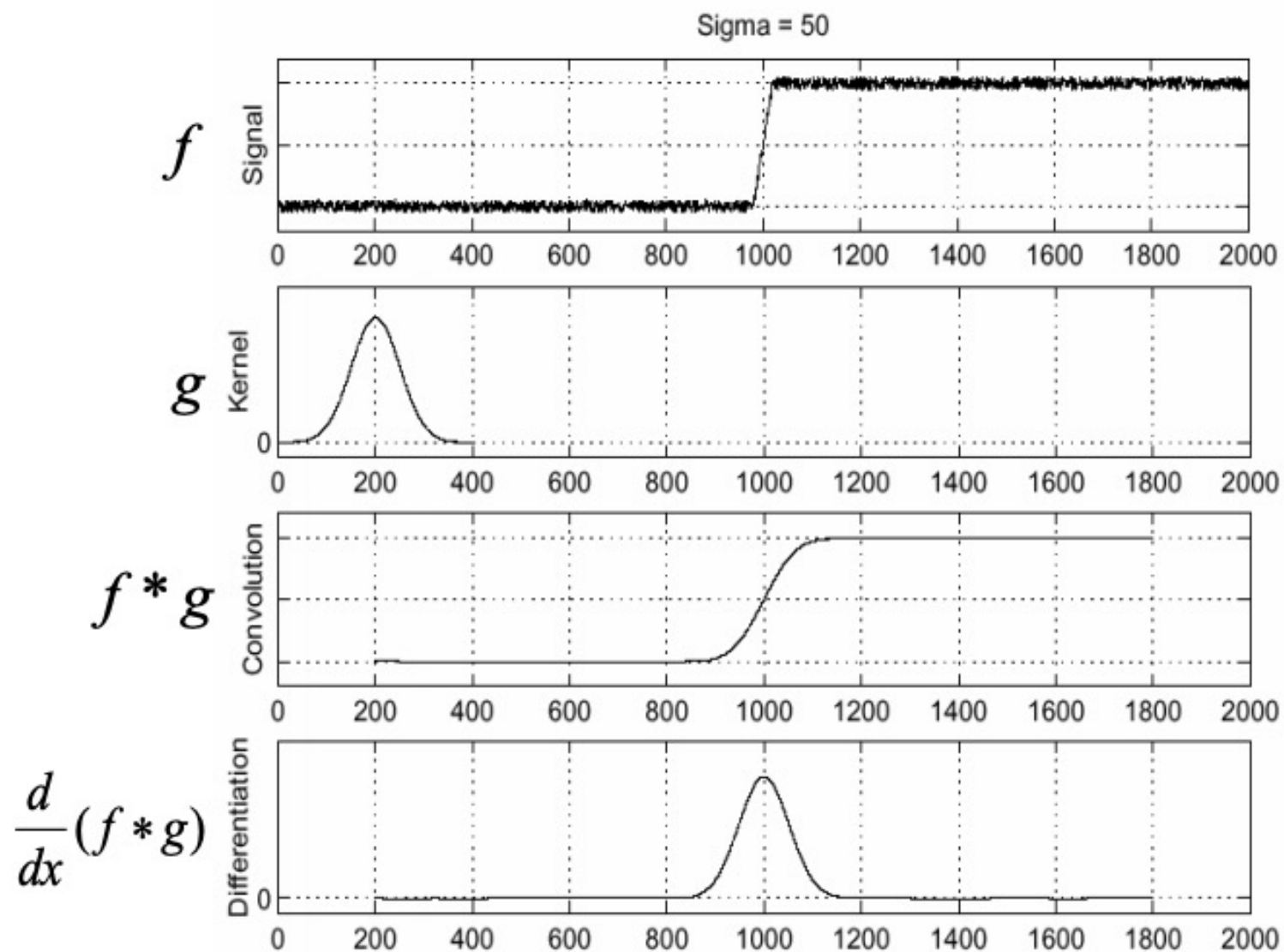
$$\frac{d}{dx}f(x)$$



Where is the edge?



Solution: Smooth First



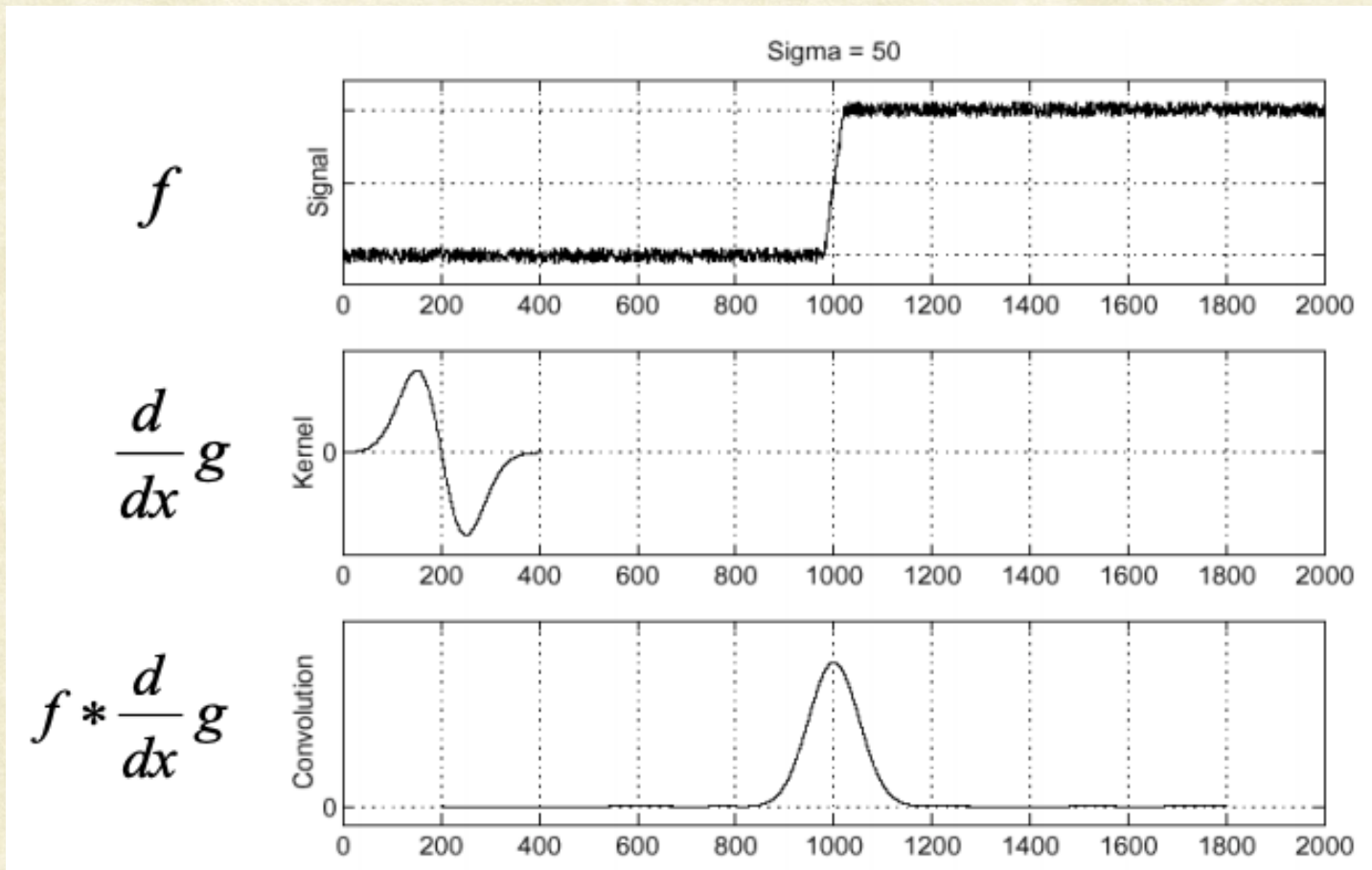


Derivative Theorem of Convolution

- Differentiation is convolution and convolution is associative:

$$\frac{d}{dx}(g * f) = \frac{d}{dx}g * f$$

- This saves us one operation:





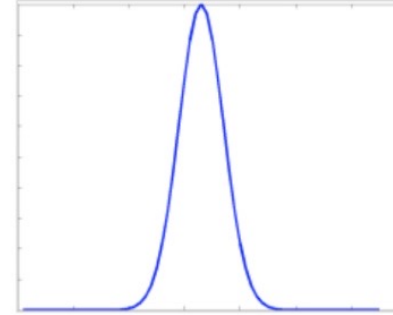
Other Important Filters

- Laplacian of Gaussian (LoG)

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CSE486

1D Gaussian and Derivatives

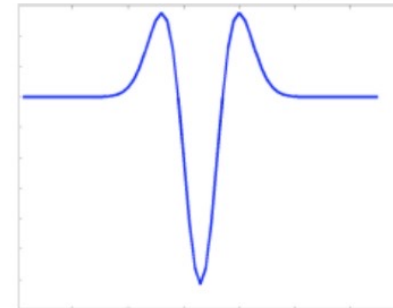
$$g(x) = e^{-\frac{x^2}{2\sigma^2}}$$



$$g'(x) = -\frac{1}{2\sigma^2} 2xe^{-\frac{x^2}{2\sigma^2}} = -\frac{x}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}}$$



$$g''(x) = \left(\frac{x^2}{\sigma^4} - \frac{1}{\sigma^2}\right) e^{-\frac{x^2}{2\sigma^2}}$$





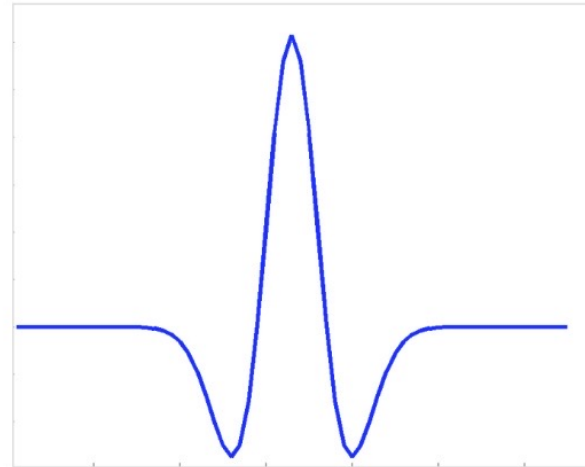
Other Important Filters

- Laplacian of Gaussian (in 2D)

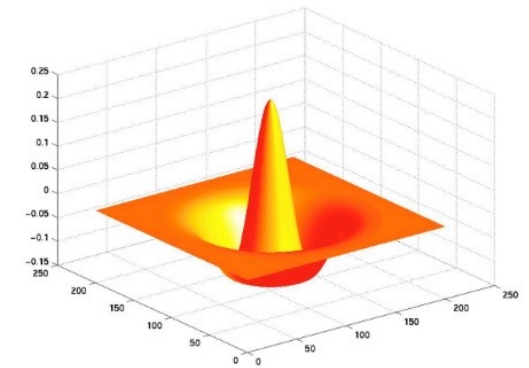
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CSE486

Second Derivative of a Gaussian

$$g''(x) = \left(\frac{x^2}{\sigma^4} - \frac{1}{\sigma^2}\right)e^{-\frac{x^2}{2\sigma^2}}$$



2D
analog



LoG "Mexican Hat"



Other Important Filters

- Laplacian of Gaussian (in 2D)
- Difference of Gaussian (DoG)

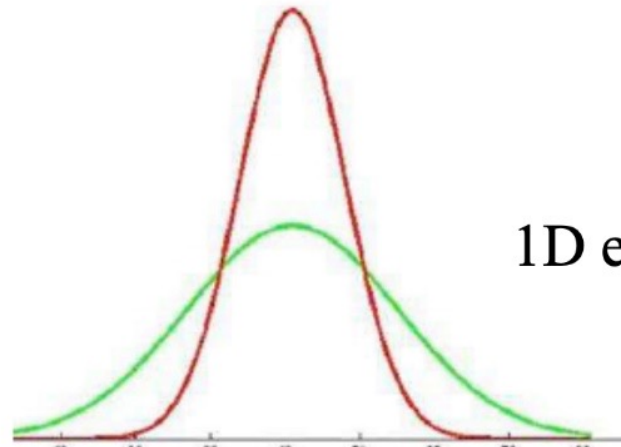
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CSE486

Efficient Implementation Approximating LoG with DoG

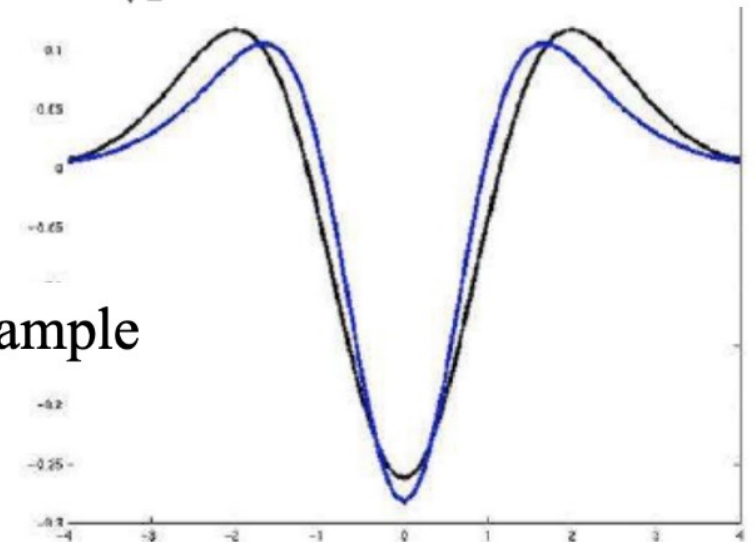
LoG can be approximate by a Difference of two Gaussians (DoG) at different scales

$$\nabla^2 G_\sigma \approx G_{\sigma_1} - G_{\sigma_2}$$

Best approximation when:
 $\sigma_1 = \frac{\sigma}{\sqrt{2}}, \sigma_2 = \sqrt{2}\sigma$



1D example





Questions?